Training Manual for Rice Production in Nigeria

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HOW TO USE THIS MANUAL

This training guide is written for extension workers at different levels of experience. They can use it to train the community knowledge workers/ community-based facilitators, lead farmers, and farmers.

Before the session

Be well organised and read through the training guide before starting a session. It will help you to feel confident. Check that you understand what to do. Organise all the necessary materials for the session like the hand-outs, flipcharts, charts, chairs, and training venue.

Beginning the Session

Greet and welcome the participants. Ask questions about the last session. For example, you can ask:

- What did we learn in the last session?
- What did you apply in your farm that you learned from previous sessions?
- What did you tell your family or community members about the session?

During the Session

Remember to speak audibly and be clear. Pause to allow time for the group to ask you questions. If farmers cannot see the board or picture during an activity, explain what is shown.

During group work, provide the farmers with flipcharts and markers. Write the participants' suggestions, answers and ideas on a flipchart and supplement with the content provided in the guide. Remind your audience that they can ask you questions if they do not understand something.

Ending the Session

1. Check what your audience learned using the questions in the question and answer activity.
2. Thank your audience for participating.
3. After each module, ask yourself the following questions and write down your answers:

• What went well in the session and why?
• What would I do differently next time?
• Which part of the session did the learners seem most interested in and why?
• Which part of the session did the learners seem least interested in and why?
• Did you manage time well? If not, why?

Reflect on each session in order to find out where you can make changes to improve later sessions.
This manual has been designed to allow readers to select and review specific sections that are relevant to their interests. Therefore, each module and sub-module are self-contained and can be read on its own. Because the focus of the guide is on how to successfully cultivate rice in the field. The manual is divided into the following modules:

Table of Contents

Module 1

Agronomic Practices in Rice Production

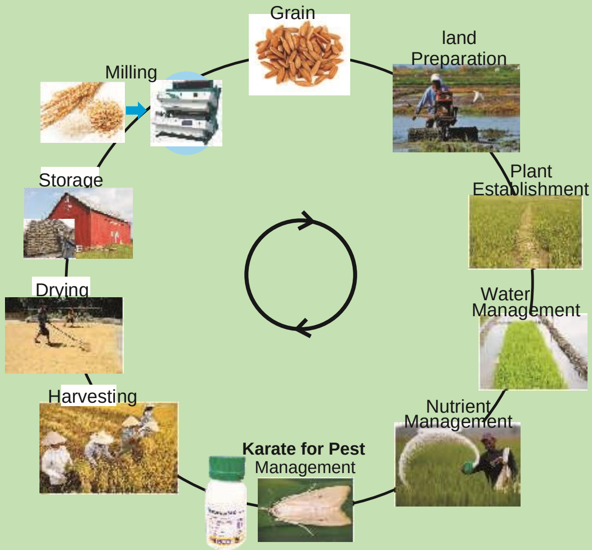
Key points in this module:
1. Basic steps in rice production
2. Training procedure

Basically, rice production entails the steps outlined below. It starts with the selection of variety, preparation of suitable land, establishment of the plant through seeding, followed by natural resource management (water, fertilizer, pests, weeds), harvesting, drying and storage or milling for food purposes.

Basic steps in rice production

The major steps:

1. Site selection
2. Selection of suitable rice variety to plant on the selected site
3. Land preparation
4. Planting (where rice is direct seeds) or preparation and management of nursery (where rice is transplanted).
5. Transplanting
6. Fertilizer application
7. Weed management
8. Pest and diseases management
9. Harvesting



In this module, these steps will be followed step-by-step and recommendations at each level will be outlined.

Training Procedure:

1. First know what existing knowledge the trainees know about rice production.
2. Identify what is the gap between what the learners know and what they need to know.
3. As the local training facilitator, initiate a discussion among the farmers about their rice production practices. Noting what are the differences between their practices and the recommended best management practices.
4. Also note innovative practices used among the farmers to overcome certain production constraints.
5. Together with farmers, facilitator and the trainer discuss what, why and when should the required changes or improvement of practices need to be applied.
6. Lead farmer should note important decision agreed on at each level of practice to enable refreshment even after training.

Module 2

Varietal Selection

Key points in this module:

1. Varieties of rice
2. Seed dormancy
3. Seed viability testing and seed requirement
4. Seed treatments

Varietal selection

1. Select variety that is adapted to local conditions.
2. The variety should be high yielding and giving desired maturity period.
3. The variety should be resistant to major pests, diseases, and soil problems in possible area of cropping.
4. It should be of good market and coking/eating quality.
5. Preferably certified seeds.

Things to note

- Rice varieties are ecology specific
- Select varieties that have market demand
- Select from current improved rice varieties that have market value.
- Research has shown that with good management, up to 3–7 t/ ha of paddy yield can be obtained with some of the recommended rice varieties.
- The recommended varieties also produce more tillers and compete better with weeds than the farmers' varieties.

Lowland:

FARO 44: Early maturing (95-110) days, for shallow swamps and irrigated ecology. Long grain. Potential yield of 7 t/ha.

FARO 52: Medium maturity (120-135) medium deep swamps and irrigated ecology. Long grain. P/Yield of 6 t/ha

FARO 61 Early maturing (100-115) days. For shallow swamps and irrigated ecology. Long grain. P/yield of 5 t/ha

FARO 57: Medium maturity (120-135), medium deep swamps and irrigated ecology.

Upland:

FARO 46: Early maturity (100-110 days)

FARO 55: Early maturity (100-110 days)

FARO 56: Early maturity (100-110 days)

FARO 53: 48 and 49; medium maturity (120-125 days)

FARO 58: Early maturity (95-100 days NERICA 7)

FARO 59: Early maturity (95-100 days NERICA 8). Potential yield of 3 t/ha

Seed dormancy

- Dormancy is the failure of good quality mature seeds to germinate under favorable conditions.
- Dormancy of freshly harvested seed should be broken by using heat treatment at 50°C in an oven if available or by placing the seeds on a plastic sheet and covering with itself or another under direct sunlight for 1 or 2 days.

- Acid treatment may also be used: soak seeds for 16 to 24 hours in 6 ml of concentrated nitric acid (69% HNO_3) per liter of water for every 1 kg of newly-harvested seeds.
- After soaking, drain acid solution off and sun-dry the seeds for 3 to 5 days to a moisture content of 14%. Store in dry conditions for sowing.
- Conduct germination test on seeds to establish rates to use based on seed viability.

Seed viability testing and seed requirement

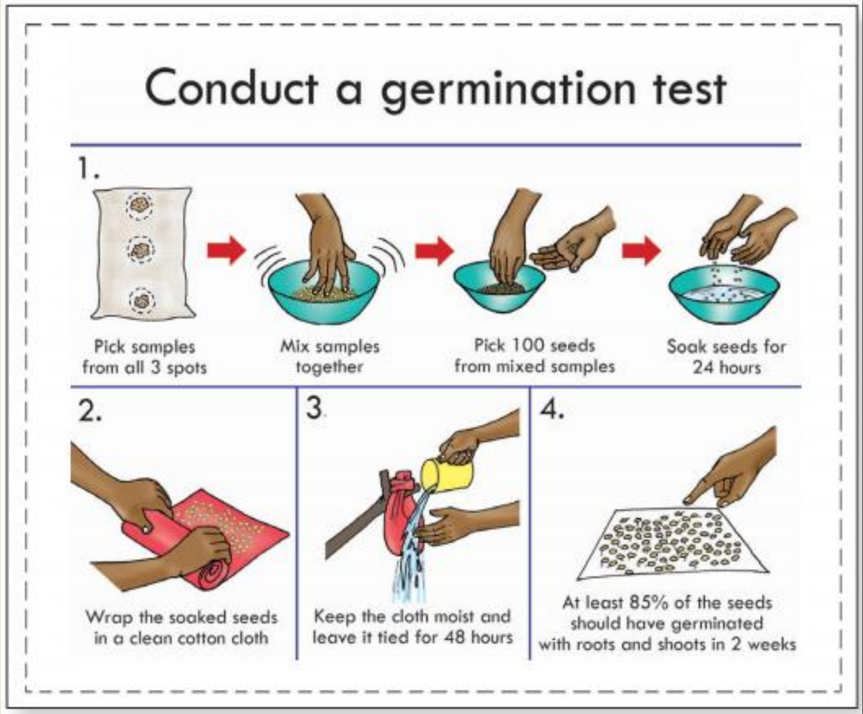
- Use only filled grains of good quality for sowing: add water to seeds and discard all empty grains that float in water.
- When the seed viability is not known, carry out a simple seed viability test to guide the actual seeds required for sowing.
- Place moistened tissue paper in a dish with lid (clean stainless dish is good). Cover and keep the dish at room temperature for 4-5 days to allow germination.
- Then count the number of sprouting seeds (only those with shoots >1 cm). If 75 germinating seeds are counted, it means the viability rate is 75% (% germination).
- If the seed rate is 80 kg/ha, the actual quantity of seeds to be used for sowing is calculated thus:

Seed required (kg/ha) = $\frac{\text{Seed rate (kg/ha)} \times \text{Area to be planted}}{\text{\% germination}}$ x % filled grain

= $\frac{80\ kg/ha\ x\ 1\ ha}{0.75}$ x 1 (assuming 100% filled grains)

= 107 kg/ha

- To estimate percentage filled grains, select at random 100 seeds from the seed lots to be sown and count the number of filled grains. If 100, it means 100% filled grains.



Seed Treatments to be considered

- First, treat selected seeds with a mixture of insecticide and fungicide.
- For example, Apron Star™ 42 WS (thiamethoxam 20 g/l + difenoconazole 2 g/l + metalaxyl-m 20 g/l) at the rate of one sachet per 4 kg seeds or any available seed dressing chemical before sowing.
- Other products can be used:
- PROCOT 40 WS (carbosulfan + carbendazim + metalaxyl-m.
- CALTHIO C 50 WS (thiram + chlorpyriphos-ethyl).
- In areas with termite and nematode problems, incorporate carbofuran (Furadan™) at the rate of 2.5 kg a.i. per hectare into planting row.
- To ensure uniform application, mix Furadan with sand at a ratio of 1:4.

- Soak seeds in water for 24 hours and incubate for 48 hours before sowing to ensure uniform seedling emergence, maturation period and good establishment.

Summary

Rice varieties are ecology specific

Use improved varieties

Break seed dormancy before sowing

Test seeds for viability

Treat seeds before sowing

Question and Answer Session

1. What are the important considerations before selecting variety?
2. What is the best variety for your location?
3. How do you break seed dormancy?
4. How do you calculate the amount of seeds required for a field?
5. How and why do you treat seeds?

MODULE 3

Site Selection

Key points in this module:
1. Considerations for site selection
2. Soil test
3. Land preparation

Because rice is water and nutrient loving crop, it is best cultivated in a location and land that is fairly fertile and nearly level, minimizing the number of water retaining barriers or levees. Some slopes are also required to facilitate adequate drainage. It is also important that the land be fertile to reduce fertilizer requirement. Rice is grown in four major ecologies, this are irrigated ecology, rain-lowland, rain-fed upland and flood prone ecology.

Important things to note

Soil puddling using a tractor-drawn rotorvator. Photo credit: website: https://www.quora.com/What-is-meant-by-puddling-inagriculture.

1. Avoid drought prone areas.
2. Avoid land with very deep ditches.
3. Choose gentle slopping land for easy drainage.
4. Fairly flat land for ease of water distribution.
5. Select naturally fertile bandwidth moderate organic matter.
6. Avoid highly iron-toxic soils.
7. Choose fertile land with good water retention capacity containing clay and organic matter.
8. Heavy soils or valleys and Fadama are preferred.

Soil Test

It is desirable to check the fertility of your soils every few years (3 - 4 years) for pH levels (how acid or alkaline your soil is), organic matter and plant nutrients. To do so, take a representative soil sample from the field after you have evaluated it in terms of topography, soil texture, drainage, color of topsoil and past management. Where the above features are uniform throughout the field, take a composite soil sample to represent the soil in the field. Where differences are large, then take one composite soil sample from each of reasonably uniform area of the field. The procedure to collect soil samples for analysis is as follows:

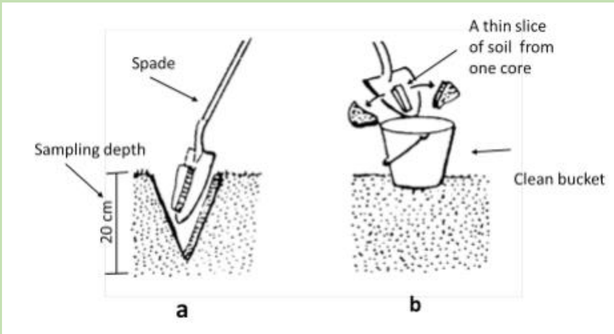


Illustration on how to take a soil sample at a specified depth (a) and a thin slice of one core sample transferred to a clean bucket (b). Figure credit: Frederick Baijukya/IITA.

i. Dig a hole 15-30 cm deep in such a way that a “V” shaped hole is made, cut out a thin slice (approximately 2.5-5.0 cm) and put it in a clean container.
ii. Continue doing the same until you have whole field sampled.
iii. After all the sample cores are taken, mixed the soil thoroughly, and remove any stones or debris.
iv. Dry the soil samples on the clean surface under the shade.

v. In the field, if available, soil testing kit can be used for rapid tests of some soil properties including pH and nutrients (N, P and K) levels.

vi. If test kits are not available, take about 0.5-1 kg of separate soil sample, label it by date of sampling, the farm name and send to the recommended agricultural service laboratories for analysis in order to inform on nutrient deficiency.

vii. Always consult the local agricultural extension agent for advice on when, how and where to analyze your soil samples.

Summary

Rice is a water loving crop
Test the soil every few years
Lower water levels lead to loss of applied nitrogen fertilizers

Question and Answer Session

1. What are the important things to note in site selection?
2. Why are soil tests important?

MODULE 4

Rice planting

Key points in this module:

1. Direct sowing
2. Transplanting
3. Nursery management

Direct sowing

Dry or pre-sprouted well-prepared seeds are directly sown by dibbling in dry soil or moist seedbed. Holes (not more than 3 cm deep) are dug in the soil at a spacing of 20 cm x 20 cm.

This spacing is achieved by the use of the marked rope. Three to 4 seeds are put in each hole and then covered by a layer of soil. If the soil moisture is adequate seedlings will emerge within 7 days after sowing. All other standard crop management activities in the field will thereafter be undertaken until the crop mature.

Bunds are made around leveled plots for properwater management. Credit: Charles Chuwa/TARI-Dakawa.

Transplanting method

Strong and healthy rice seedlings should be transplanted on a well-prepared seedbed, normally after the incorporation of basal fertilizers. To produce strong and healthy seedlings, the well- prepared, pre-sprouted seeds of the selected variety should be sown in the nursery preferably a few days before preparation of the main field.

Nursery preparation and management

The nursery should be located as close as possible to the main field. The site should preferably have fertile soil and adequate water and be well-protected from wild/farm animals.

The following activities is to be done:

- Making seed beds of convenient sizes.
- Puddling the field and leveling it.
- Dividing the field into 1 x 10 m nursery beds.
- Making furrows around each nursery bed.
- Doing final leveling of nursery beds.

Soil preparation for dry direct-seeded rice (left) and for traditional transplanted-flooded rice (right) Credits: INRA and Springer-Verlag France 2014

Sowing and managing seedlings in the nursery

- Broadcasting (evenly) pre-sprouted seeds on nursery beds and tapping the seeds into the soil (400 g or at about one handful of seeds per 1 m^2).
- Keep the soil moist/wet until one week after seedling emergence.
- Gradually raise the water level, from 1 to around 5 cm (depending on seedling height) until transplanting.
- Keep the nursery beds free from weeds.
- Apply pesticides to manage diseases and insect pests where necessary.

Some consideration for transplanting

Transplanting must be done in a well-prepared field (leveled and surrounded by bunds). In the field, the facilitator should ensure full participation of all participants in performing the following activities at least one day before transplanting seedlings:

- Introducing water into the field at shallow depth (around 3 cm if possible) or just to wet the soil.
- Broadcasting basal fertilizers (DAP, MOP, or manure) at recommended rates.
- Properly incorporating the basal fertilizers into the soil by final puddling (using rotorvator, hand hoe, or ox-plow).
- Performing final leveling of the field using rakes or any other suitable implements.

Seedling preparation and planting

Farmers should note the proper age for transplanting seedling. Seedlings should be transplanted when they are old enough to be conveniently handled and not too old to produce many strong tillers. Preferably seedlings should be 15-30 days old at transplanting.

Well established rice field. Photo Credit: Stock Images

Some considerations on how to prepare seedlings for transplanting

Seedlings should be uprooted and transplanted on the same day. Seedling preparation is achieved by the following procedure:

- Prior to pulling out seedlings, allow enough quantity of water into the nursery beds.
- Select quality seedlings with uniform height.
- Grasp seedlings near the base and uproot at least three at a time.
- Gently uproot the seedlings at an angle of about 30 degrees from the horizontal.
- Wash the seedlings thoroughly to remove soil adhering on roots.
- Bundle the seedlings to facilitate transportation and handling during planting.
- Place the seedlings in water during planting to prevent wilting

Transplanting

After the soil has settled in the puddled and well-levelled field, 2 to 3 seedlings should be planted per hill at a depth of about 3 cm and spaced at 20 cm between rows and 20 cm within a row. The use of the marked transplanting rope enables spacing to be kept constant throughout the field. The spacing may be increased with decreasing soil fertility and water supply, and vice versa. After transplanting of the whole field is complete, collect all remaining seedling bundles, untie them and plant them in one alley for future gap-filling to be accomplished within 10 days.

Line transplanting using a marked rope. Credit: Charles Chuwa TARI-Dakawa.

Summary

Use a spacing of 20cm x 20cm
Nursery should be sited close to the main farm
Keep nursery free from weeds
Fields for sowing should be levelled

Question and Answer Session

1. How is direct sowing carried out?
2. How do you prepare your nursery?
3. How do you manage seedlings in the nursery?
4. Why should you transplant?

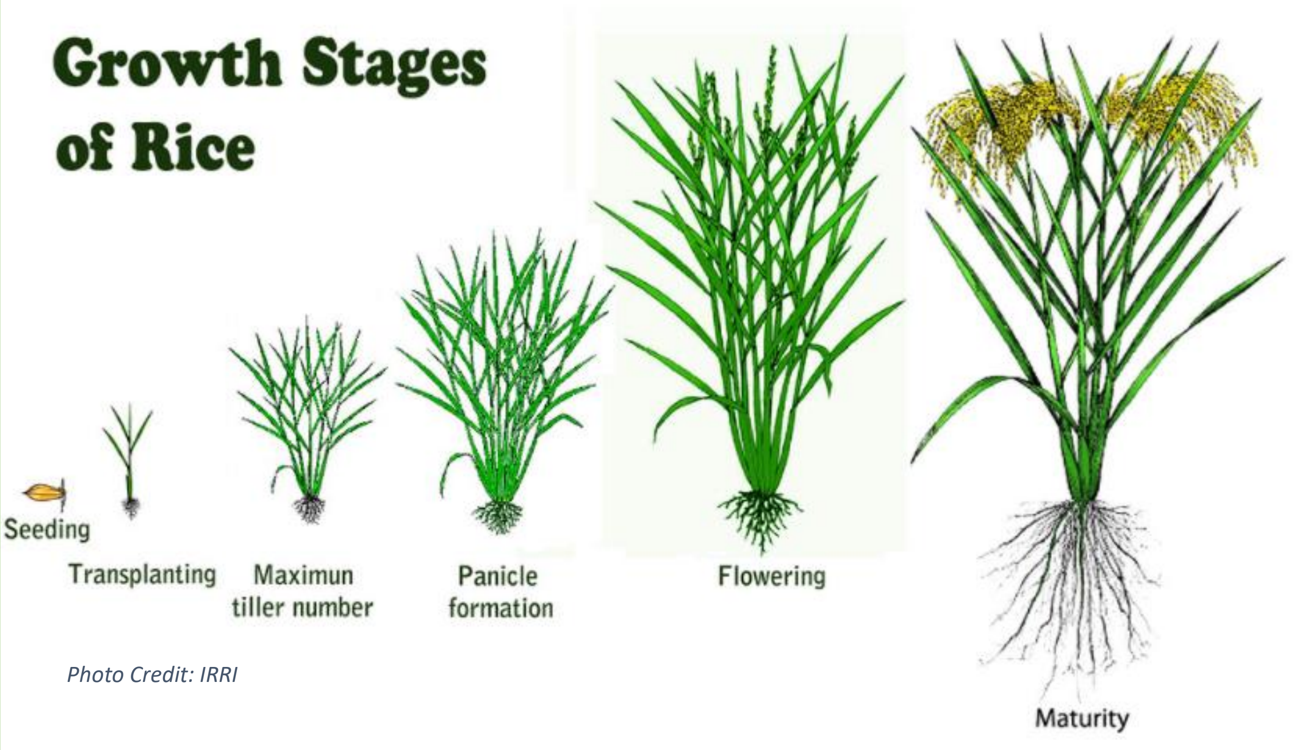
MODULE 5

Field and water management

Key points in this module:
1. Rice growth stages
2. Water management

Growth phases and stages of the rice plant

The rice plants pass through three major growth phases from seedling emergence to maturity. These are the vegetative phase (number of days from seedling emergence to panicle initiation), reproductive phase (from panicle initiation to heading), and ripening phase (from heading to maturity). The length of the vegetative phase differs among varieties, but the reproductive and maturing phases are equal for all varieties.



Sowing to mature seedlings	Transplanting to full establishment	Tillering	Internode elongation	Panicle initiation to booting	Heading and flowering	Grain filling to maturity
Vegetative phase 35-55 days (short-duration varieties: 100-120 days) 55-75 days (medium-duration varieties: 120-140 days) 75-95 days (long-duration varieties: > 140 days)				Reproductive phase 35 days		Maturing phase 30 days

Water management

Water availability largely determines the potential crop yield. Good water control increases crop yields and grain quality as well as improving the use efficiency of other inputs such as fertilizer, herbicides, and pesticides. To maximize water -use efficiency:

- Maintain the bunds
- Level the fields
- Puddle the fields where possible

- Use direct-seeding techniques
- Use short-duration crops
- Harvest on time

Key points to consider after crop establishment

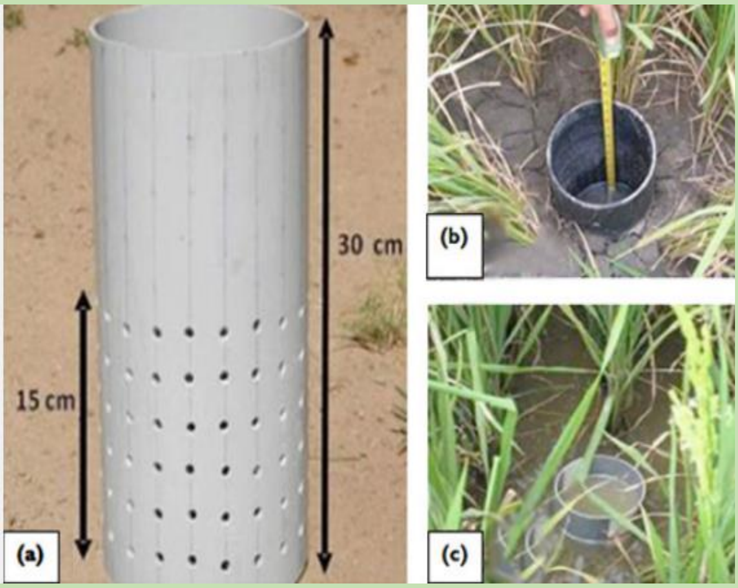
- Continuous flooding of water generally provides the best growth environment for rice.
- After transplanting, water levels should be around 3 cm initially and gradually increase to 5−10 cm (with increasing plant height) and remain at this level until the field is drained 7−10 days before harvest.
- For direct wet-seeded rice, the field should be flooded only once the plants are large enough to withstand shallow flooding (3-4 leaf stage).
- Lowland rice is extremely sensitive to water shortage (below saturation) at the flowering stage. Drought at flowering results in yield loss from increased spikelet sterility, thus fewer grains.
- Keep the water level in the fields at a minimum of 5 cm at all times from heading to the end of flowering.
- In case of water scarcity, apply water-saving technologies such as Safe Alternate Wetting and Drying (AWD) and consider changing the planting method from puddled transplanting to non-puddled transplanting or dry direct-seeding.

What is AWD?

Alternate Wetting and Drying (AWD) is a water-saving technology that lowland (paddy) rice farmers can apply to reduce their water use in irrigated fields. In AWD, irrigation water is applied to flood the field a certain number of days after the disappearance of ponded water.

A practical way to implement AWD is to monitor the depth of ponded water on the field using a Field Water Tube.

- After irrigation until 5 cm of water is ponded above the soil surface, the depth of ponded water will gradually decrease.
- When the ponded water has dropped to 15 cm below the surface of the soil, irrigation should be applied to re-flood the field with 5 cm of ponded water.
- From one week before flowering to one week afterwards, ponded water should always be kept at 5 cm depth.
- After flowering, during grain filling and ripening, the water level can drop again to 15 cm below the surface before re-irrigation.
- AWD can be started a few days after transplanting (or with a 10-cm tall crop in direct seeding).



The diagram of a 30-cm Field Water Tube perforated to 15cm (a) being inserted into the soil at a designated depth prior to flooding (b) and after flooding (c). Photo credit: Charles Chuwa /TARI-Dakawa.

- When many weeds are present, AWD can be postponed for 2-3 weeks until weeds have been suppressed by the ponded water.
- Local fertilizer recommendations as for flooded rice can be used. Apply fertilizer N preferably on dry soil just before irrigation.

The field water tube

- A tube can be made of 30-cm long plastic pipe or bamboo and have a diameter of 15 cm or more so that the water table is easily visible. Perforate the tube with holes on all sides..
- Dig the tube in the soil so that 5 cm protrudes above the soil surface.
- Take care not to penetrate through the bottom of the plow pan. Remove the soil from the inside so that the bottom in the tube is visible at 15 cm depth.
- check that the water level is the same inside and outside the tube.
- The tube should be placed in a level part of the field close to a bund, so it is easy to monitor the ponded water depth.

Summary

Major growth stages of rice are vegetative, reproductive and ripening phases
Water management is the key component that determines rice yield
Alternate wetting and drying helps in water retention in the field

Question and Answer Session

1. What are the growth stages of rice?
2. When is the vegetative phase?
3. How do you manage water in your field?
4. How do you make a field water tube?

MODULE 6

Soil fertility and weeding management

Key points in this module:

1. Fertilizer application
2. Weed management
3. Spraying precautions

Fertilizer Application

There are different ways to replenish nutrients in the soil. The major ones are as follows:

- Harvesting only the economic part of the crop, leaving the rest in the field to decompose.
- Returning all the residues to the field after harvesting.
- Applying organic manures; green manure, farmyard manure, compost, etc.
- Applying industrial/chemical fertilizers.
- Out of the four ways above, the use of industrial fertilizers is inevitable because of its quick response when applied; it gives out the required nutrients on time owing to its quick solubility. Organic materials respond very slowly because of their low rates of decomposition when applied to the field.

In situations where it is not possible to conduct a soil test owing to the high cost and unavailability of analytical services, or when the farmer is running out of time because the crop is subnormal in growth, the general recommendation is to apply 120 kg N, 40 kg P2O5, and 40 kg of K2O/ha.

That means basal application of 4 bags of NPK/ha before planting and top dressing of 1 bag of Urea at 3 to 4 weeks after seeding or transplanting and one bag at 6-7weeks after seeding or transplanting. Broadcasting is the best method for rice paddies.

Things to note:

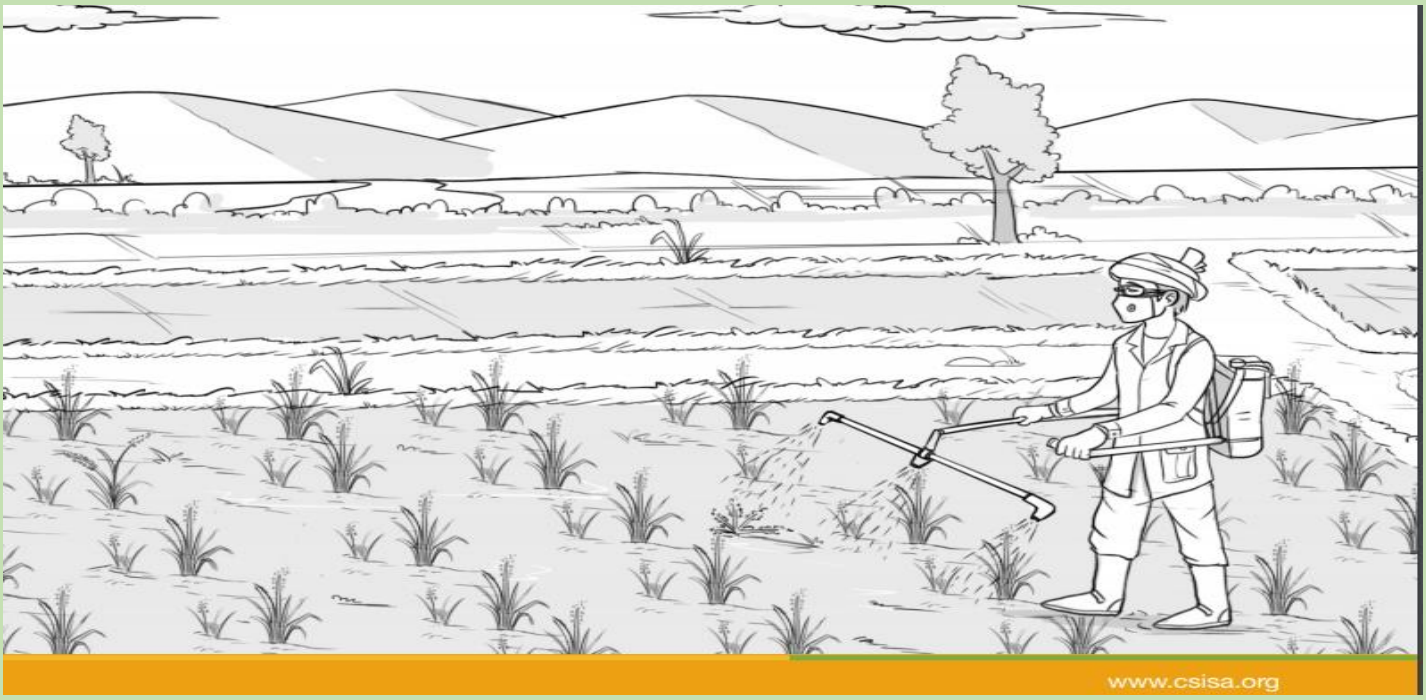
- When applying fertilizer, reduce water level to 2-3 cm, just enough to dissolve the fertilizer. Do not dry the field.
- Use the leaf color chart (LCC) if available.
- Side dress 1/3 of N fertilizer 5-7 days before panicle initiation. Apply at 40-50 days after sowing (DAS) for early maturing varieties and 55-60 DAS for medium maturing varieties.
- Do not top-dress while the leaves are wet. The fertilizer will stick to the leaves and may cause leaf burn. Dissolved fertilizer will be lost in the air when the droplets dry.
- Do not top-dress where there is impending heavy rain because the fertilizer will be washed out in the field.
- Keep the fields free from the weeds. Weed before fertilizer application.

Weed management methods and control with herbicides in rice seed production as Post-emergence

Timely control of weeds is critical for high yield and quality, it is therefore important, that:

- Weed management start even before sowing or transplanting, depending on the level of weed infestation, first weeding is done 2-3 weeks after sowing with a second weeding at 4-5 weeks, and a third weeding at 7-8 weeks, depending on the duration to maturity of the crop and the level of weed infestation. Alternatively, apply pre-emergence herbicides such as glyphosate and sow 2 weeks later, or spray with gramoxone, which is another pre-emergence herbicide and sow 1-2 days later.

- Remember, the use of herbicides in weed management requires some knowledge of the right product for a particular crop, the right time to apply it using the correct dilution. Pay attention to the instructions in the use of the chemicals. Seek advice from your extension agent on appropriate herbicide types and rates if



you are not conversant with their use. Also ensure that the farm is weed-free at all times to get good quality grain that are well filled and will perform vigorously from planting maturity.

For transplanted rice

- Apply pre-emergence herbicide (e.g., pretilachlor or butachlor at 2−3 DAT)
- If grass weeds are the main weed problem, apply early post-emergence herbicide
- Do not allow soil surface to dry after transplanting. Keep the soil moist to saturated. Dry soil reduces the performance of pre-emergence herbicides.
- Maintain a 5−7 cm water depth to prevent germination of weeds until 7−10 days before harvest.
- If herbicides have not been applied, or if weeds are emerging, you may use push weeder to control weed seedlings that are at 3−4 leaf stages. Irrigate one day later to prevent buried and uprooted weeds from recovering.
- Maintain shallow flooding 7−10 DAT, drain the field, then push the weeder down the row to bury emerged weed seedlings.
- Leave the field saturated for 2 days to keep the buried weed seedling in the mud layer then flood the field up to 5 cm of water.
- Hand weed as needed until the canopy closes.

For wet seeded rice (broadcast)

- Apply pre-emergence herbicide (e.g., pretilachlor + fenclorim 2−3 DAS).
- If grass weeds are the main weed problem, apply early post-emergence herbicide

- For post-emergence herbicide application, drain water in the field to expose weeds, then spray the herbicide.
- **Note:** Post-emergence herbicide should come in contact with leaves of weeds to be absorbed by the weeds. When weeds are submerged in water, post-emergence herbicide will not be effective.
- Do not allow soil surface to dry after seeding. Flush irrigate as needed to keep the soil moist to saturated. A dried soil surface will reduce the performance of pre-emergence herbicides. Irrigating more than 10 days after seeding encourages more weed growth and deeper water level is needed to control weeds.
- If herbicides have not been applied, or if weeds are emerging, you may use push weeder in a row-seeded crop to control weed seedlings that are at 3−4 leaf stages. Irrigate one day later to prevent buried and uprooted weeds from recovering.
- Maintain a 5−7 cm water depth to prevent germination of weeds until 7−10 days before harvest.
- Hand weed as needed until the canopy closes.

Above: Different types of sprayers (Knapsack and Mounted)

Precautions while spraying in the field

- Read the label on the package before using.
- Only apply chemicals obtained from recognized sources.
- Adhere to manufacturer's instructions and recommendations on the label.
- Use correct kind, dosage and application technique and appropriate safety precautions.
- Determine calibration rate and go by it.
- Seek technical advice when in doubt.
- Do not use expired agrochemicals.
- Wash your hands thoroughly with soap and water after using agro-chemicals.
- Do not eat while applying agro-chemicals.
- Dispose agrochemicals containers in a secure place.

- Wear protective equipment while you apply to ensure safety. You must wear goggles, nose mask, overalls, gloves and closed gumboots.

If the herbicide accidentally touches the skin and sensitive organs such as the eyes, wash the affected part with soap and clean water. If you feel some pain seek medical attention

Summary

Fertilizer helps improve yield and replenish the soil nutrients
Use suitable fertilizer for your rice fields
Timely control of weeds is critical for high yield and quality
Herbicides should come in contact with weeds to be effective
Precautions should be taken when handling herbicides

Question and Answer Session

1. What are the different ways to replenish the soil nutrients?
2. What are the things to note when applying fertilizers?
3. How do you control weeds in your rice paddy?
4. What are the precautions to note when spraying the field?

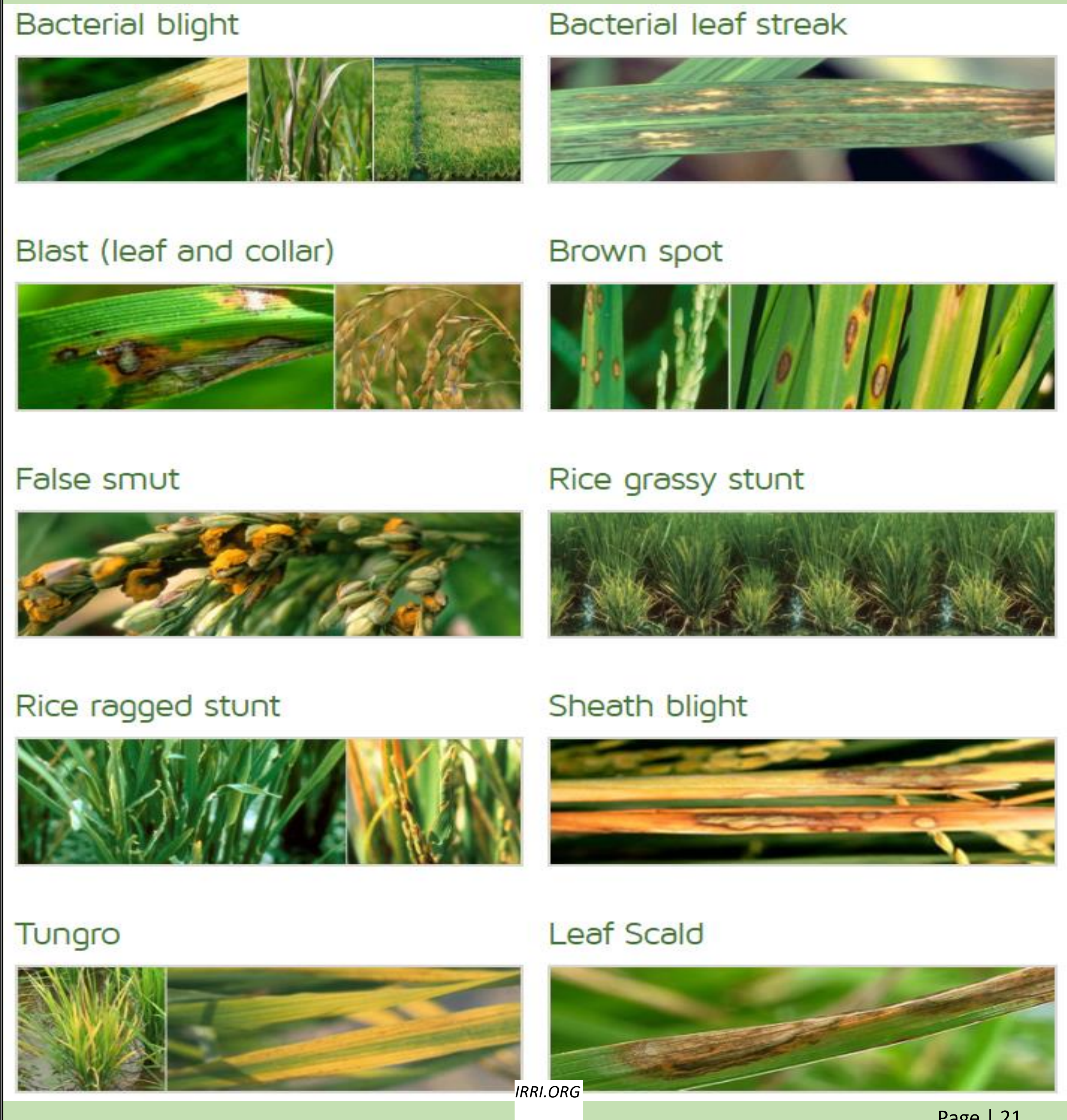
MODULE 7

Pest and Disease Management In Rice Production

Key points in this module:

1. Diseases
2. Insects pests
3. Recommended Management Techniques

Diseases: Disease damage to rice can greatly reduce yield. They are mainly caused by bacteria, viruses, or fungi. Planting a resistant variety is the simplest and, often, the most cost-effective management for diseases.



Narrow brown spot

Red stripe

Bakanae

Sheath rot

Blast (node and neck)

Stem rot

Bacterial sheath brown rot

Rice stripe virus disease

IRRI.ORG

Insects pests

In the early stages of the rice crop, several common insects such as the leaf folder, whorl maggot, and armyworms can cause highly visible damage symptoms; however, the damage is rarely enough to reduce yield because the crop can compensate for early damage over the rest of the growing season.

Armyworm

Green semilooper

Greenhorned caterpillar

Rice bug

IRRI.ORG

Black bug

Zigzag leafhopper

Rice skipper

Rice thrips

Rice whorl maggot

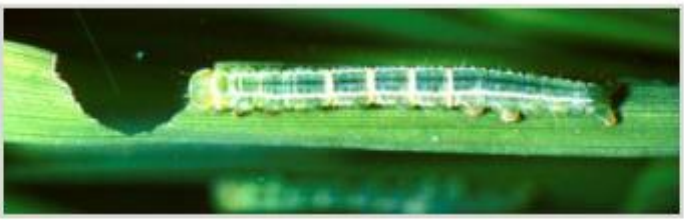
Mealy bug

Mole cricket

Ant

Armyworm

Green semilooper



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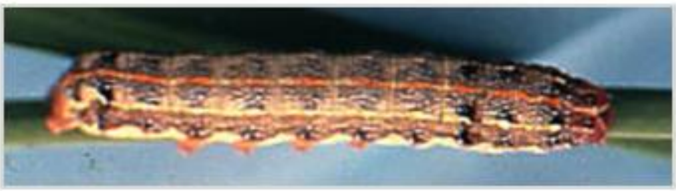
Planthopper

Field cricket

Cutworm



Green leafhopper

Rice caseworm

Grasshopper (Short-horned) and Locust

Rice gall midge

Rice hispa

Stem borer

Root grub

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In most cases, insecticides applied in rice fields during the early crop stages to control leaf folders or whorl maggots are unlikely to benefit farmers economically. Instead, they can cause an imbalance in the natural insect population that may lead to pest outbreaks. Later stages of spraying is recommended.

Recommended Management Techniques

- Adhering to best practices in pest management can reduce any undesirable side effect and hazards to a minimum level.
- These could be applied by selecting a combination of the most efficient and cost-effective technique available (integrated pest management).
- The choice of these integrated measures will depend on economic, type and size of farm, conditions prevailing in the area and the future use of the commodities.
- Synchronized planting can evade pest infestation. During the early growth stage of the crop (30-40 DAT), it is not necessary to spray pesticides against leaf feeding insects, as the rice crop can compensate for leaf damage.

Cultural Control

- Plant early
- Plant early maturing varieties
- Plant simultaneously in a district to reduce the population of certain insects.
- Flood the field to reduce the population of some insect pests e.g. armyworms. -Weed at appropriate time.
- Burn straws and stubbles and plough into the soil.

Varietal Resistance

- Plant rice varieties that inherently attract relatively fever insect species and suffer less damage.

Biological Control

- Rice insect pests are attacked by numerous parasites, predators and pathogens.
- Avoid use of much broad-spectrum non-selective insecticides.

Chemical Control

- Insecticides are chemical compound used to kill insects
- Use them properly, they are harmful to human beings.
- Active ingedients and the dilution of the chemical should be known before they could be applied in either.

Solid Form

- Dust or power without the need for water e.g. Aldrin
- Granules – small solid particles used for broadcast e.g Furadan.

Liquid Form

- Wet table power – a wetting agent is added so that it can stick to sprayed surface.
- Emulsifiable concentrate (EC) the active ingredient is mixed with an appropriate solvent as a carrier together with an emulsifier.

Using Herbicides

- Drain water from the field.
- Apply Glyhphosate as pre-emergence two or three weeks before seeding or transplanting at 4 liters per ha in 120 litres of water.
- Apply propanil plus 2-4 D or Orizo plus as 3-4 weeks after seeding or transplanting at 4 litres per ha.
- Hand weed again around 40 days after transplanting.

Summary

Diseases reduce yield
Insect pests attack the rice in several forms
Integrated disease and pest management helps combat the spread of pests and diseases.

Question and Answer Session

1. What diseases of rice can you identify?
2. What pests of maize do you know of?
3. What are the recommended management practices?
4. What are the forms of insecticides?

MODULE 8

Yield forecasting

A pre-harvest yield estimate can be accurate to within 5 percent of the actual harvested yield if the correct procedure is used. When working with trial and demonstration plots, you should always take such a pre-harvest yield sample of both the test plot and the control plot. There is always the chance that the plots might be inadvertently harvested before the agreed-upon time without the yields being measured. Pre-harvest yield sampling is also a quick way of estimating crop yields in farmers' fields.

General Principles of Yield Sampling

- Samples should be collected at random for various portions of the field or plot. Do not purposely select samples from higher- or lower-producing areas within the plot or your estimate may be very inaccurate A random sampling pattern should be determined before you enter the field so you will not be tempted to choose them by visual appearance.
- Don't collect yield samples more than one week before the actual harvest.
- When taking each sample, the area (or row length) to be harvested must be precisely measured. Do not estimate' Remember that any error in the sample area size will be magnified hundreds of times when converting the yield to a larger land unit basis.
- You must adjust the sample weights to account for factors like excess moisture, damage, and foreign matter.

How to Take Samples and Estimate Yields

1. The Sampling Procedure

- Number of samples: For plots less than 0.5 ha, take a minimum of five samples. For plots of over 0.5 ha, take between five and ten. If crop growth is not very uniform, take ten samples.
- Size of each sample: Take each sample from the same sized area or same amount of row length. Individual sample size should be between 2.5 and 5.0 square meters. For row crops, the area of a sample is determined by multiplying row length by row width. (Harvesting three meters of corn row planted in rows one meter wide will give you a sampling area of three-square meters.)
- Taking a random sample: Decide on the sampling pattern before entering the field, and do not deviate from it. To randomize, the field can be divided up into sections and each section given a number drawn from a hat. Or you can pick randomized starting points at the side of the field and then enter random distances from the starting point. A good system for row crops is to number the rows and select them at random, then select the distance into the row (field) at random. NOTE: Exclude three meters or four rows of perimeter from your sampling area along all four sides of the plot to ensure sampling from the heart of the plot.

2. Accuracy: Use a tape to measure each sampling area or row length. Use an accurate scale to record the total weight of the samples within one plot.

3. Handling the Samples: The samples should be harvested and processed according to local prevailing methods. If drying is required before shelling or threshing, be sure the location is secure and free from rodents or birds.

4. Weighing the Sample: Use an accurate portable scale. You do not need to weigh individual samples separately, but only the fatal collective sample from the plot. If you cannot find a good portable scale, have the grain weighed in town.

5. Checking Grade: Take a random sample of the collective sample and have it checked for moisture content and any other graded qualities if necessary. (Refer to the storage section in Chapter 7 for how to determine grain moisture content.)

6. Yield Calculations:

Size of total sample area = No of samples x size of individual sample areas

Estimated yield = weight of total sample area x $\frac{total\ size\ of\ the\ plot}{total\ sample}$

7. Correcting for Moisture: Yields are usually based on grain that is dry enough to store in shelled form (usually 13-14 percent moisture content). If you base your estimates on the weight of a high moisture sample, you should revise the yield downward using this simple formula (otherwise, dry the grain first).

Grain weight after drying = [% dry matter before drying / % dry matter after drying] x original grain weight

Example: Suppose you weigh a collective sample of "wet" grain and then estimate the plot yield to be equal to 3500 kg/ha. A moisture test shows the sample has 22 percent moisture; what is the actual yield based on 13% moisture?

22% moisture = 78% dry matter,

13% moisture = 87% dry matter

$\frac{78\%}{87\%}\ x\ 3500\ kg/ha$ = 3138 kg/ha yield based on 13% moisture

A Yield Estimate Example

Suppose you are taking a yield estimate on a farmer's rice plot which is slightly less than 0.5 hectare. The rows are planted 90cm apart, and you decide to take six samples, each consisting of 1/1000th hectare of row length. The collective weight of the shelled, dried rice is 18 kg. What is the estimated yield on a per hectare basis?

Solution:

area of collected sample = 6/1000ths of a hectare = 60 sq. Meters

$18kg\ x\ \frac{10000m2}{60m2}$ = 3000 kg/ha estimated yield

Summary

Yield forecasting helps the farmer know the amount of yield he would harvest.

Question and Answer Session

1. How do you sample a field?
2. What is the formular to calculate a yield estimate?

MODULE 9

Harvest and Post-Harvest Operations in Rice Production

Basic Learning Objectives

It is expected that at the end of all the modules trainee will be able to:

- Discover good post-harvest and processing practices.
- Exchange information on post-harvest operation and rice parboiling.
- Discover operations in the rice post-harvest that will be considered as critical control point (CCP).
- Understand production branding and packaging for rice.
- Have enhanced knowledge good practices on rice parboiling technologies.

Procedure

While organizing training using this module and for maximum impact, the following procedures are necessary.

- Discuss farmers' post-harvest practices areas where practices are in line with recommendations and where it is not.
- Facilitator or instructor launches debate on good post-harvest management and processing practices beginning with harvesting operation. Emphasizing best harvesting period, when at least 80% of the grains have turned yellow and the remaining 20% has reached dough stage; grains are clear and hard when dehulled and consequence of late harvesting: losses due to rodents, birds and shattering.
- Discuss the importance of drying harvested panicles, threshing winnowing as soon as possible after harvesting to ensure good grain quality.
- Take primary post-harvest operations one after the other and interactively discuss with farmers recommended good practice, highlighting possible area where loss may occur and possible remedy.
- After carefully reviewing the primary post-harvest operations, initiate discussion on Secondary Post-harvest Operations (parboiling, milling, packaging and marketing).
- Initiate debate on good parboiling operation, allowing each trainee to say and agree on common practices, before introducing the importance of parboiling, good practices and stepwise operations that will enhance the recovery of high head rice and low broken rice.
- Introduce cost saving operations that will make rice processing competitive such as the use of fuel-efficient vessel and stoves, agro-processing waste to fire vessels, proper cleaning to reduce foreign materials etc.
- Introduce the concept of rice packaging, its importance, and strategies for branding and marketing.
- Finally review the good practices outlined for the primary and secondary postharvest operations. This should be done by each farmer explaining what he or she has learned during the training and also highlighting what need to be changed in his usual practices.
- Encourage sharing of such information with other farmers within the same community or cooperative.
- Also help lead farmer or farmer's leaders to have all basic steps clear as he/she will step the training down at village level where extension officer is not available.

Recommended Harvest and Post-Harvest Operations

- Rice is mature when 80% of grains turned straw colour.
- If the field is still flooded, remove water and leave the crop to dry to 18 to 20% moisture content.
- Harvest and thresh if dry.

- Thresh on concrete floor or tarpaulin, or dry threshed paddy on concrete floor or machine drier to avoid stones and gravels.

Things to note

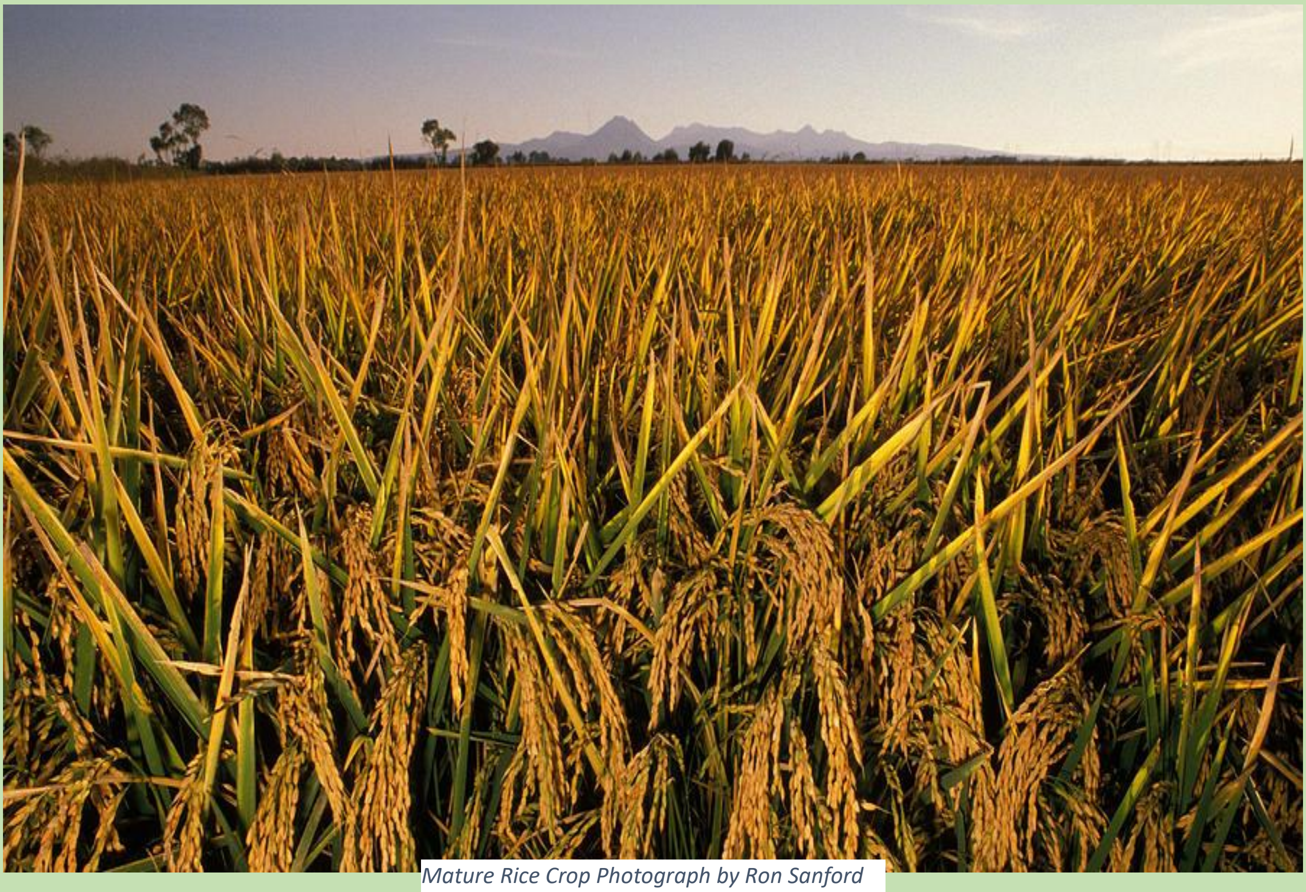
Harvest paddy when 80-85 percent of the grains are mature. Grains at the tip of the panicle must be hard and golden yellow, even while grains near the base of the panicle are less mature. Delay in harvesting will cause the grains at the tip of shatter.

Generally, draining of the field be done 2 weeks before harvest. However, depending on soil type, gradually drain the field up to saturation point, preventing abrupt drying of the soil as this will affect grain quality. Harvested crops must be threshed immediately to minimize field losses and grain quality problems.

Thresh, clean, and dry grains immediately to 14 percent moisture content, or lower before storage.

How to observe the paddy field in order to determine an optimum harvest time?

1) Observe your field to notice when the panicles, not the rice plant starts getting yellowish in appearance. Do not confuse the colour of the straw with the yellowish colour of the panicles. Yellowish colour of the straw may result from early senescence of the leaves.
2) Pay attention to the panicles and observe when individual panicles in different parts of the field, have up to 80% maturity in the upper part of the panicle.
3) Waiting for the remaining 20% to mature may lead to shattering of the upper grains, which are very often better filled, thus, resulting in yield reduction.



Mature Rice Crop Photograph by Ron Sanford

Threshing

In manual threshing several methods are available. In some places threshing is done using legs to march on the straw, others beat the straw against drums, others put the straw in bags and beat it against tree trunks, stone etc while others use sticks to beat rice straw heap.

This method is time consuming, laborious, slow and output is quite low. Contamination of paddy with sand, stone, immature grain and other foreign materials is high. There is also loss of grain leading to reduction in overall yield.

Manual Threshing

Mechanical Threshing

However, there is an intermediate technology for rice threshing that is adoptable to small farm size and with efficient output. These threshers are either semi motorized or completely motorized. These are already available in several places in sub-Saharan Africa, particularly in irrigated areas or developed low land.

The thresher provides clean paddy without damaging the grain because of its axial flow system. Straw are chopped into and discharged as waste which may be collected and preserved for animal feed or incorporated into the soil to improve soil fertility. It is easy to be operated by both men and women.

This to note

- In the use of motorized threshers, there is need to have adequate knowledge on the adjustment of the motor and air velocity.
- It provides opportunity to gather chopped straws for animal feds and no winnowing after threshing.
- Initial cost of production may be slightly high for single farmer, but cooperative may buy and use it as income generation venture.

Cleaning Manual Cleaning of paddy

In manual winnowing the threshed materials are thrown up along the directions of flow of the wind, blowing away all the chaff and lighter materials. Material of the same density as that of the rice grains are also retained. The process requires much labour, is slow and low in output.

Mechanical Paddy Cleaner

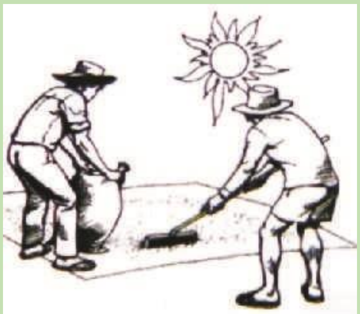
Also available are mechanical winnowing blowers and cyclone. With hand operated blowers, the operator winds the machine to generate a centripetal force which separates the grains from the chaff. The grains drop to the bottom while the chaff is blown out through the exhaust. This has a larger output than the manual winnowing and the problem of recontamination is eliminated. The motorized winnowers operate the same way but have increased output. They are however more expensive and are limited in availability. There is also the problem of the cost of fuel, maintenance and spare parts.

Things to note

1. Use a clean surface, plastic sheets, tarpaulin etc when cleaning paddy to avoid contamination with sand and other extraneous materials that may reduce the quality of rice.
2. Clean the machine after cleaning one variety before introducing a different variety, especially if rice processed for seed.
3. For efficient manual cleaning always stand where the wind speed is minimal not to allow blowing away of good grains.
4. When water cleaning is used especially during parboiling, stir properly and don't pack the bottom sediments.

Rice Drying

The major objective of rice drying is to reduce grain moisture content to a level that will meet the recommended levels for safe, long term storage and milling operations.



Depending on the ecology, rice drying in Nigeria is by direct sun light. The extent and efficiency of the operation depends on the sun and general weather conditions. Paddy dried under high intensity sunlight for a long time may develop cracks and on processing result in high level of broken grains. Rice are dried on highway sides, on mats, etc. Most of these drying facilities expose the grains to contamination and loss due to spillage, animal consumption and wind.

Parboiling

- Parboiled rice has the advantages of better storage, cooking quality, being richer in food value, devoid of unpleasant odour and breaking less during milling.
- Soak paddy in hot water at 70°C (hot enough for your fingers to withstand the heat for about two seconds) for 5 to 6 hours.
- Discard all floating grains.
- Parboil rice by steaming soaked paddy in a jute bag for 10 to 16 minutes.
- Suspend the bag over steaming water in a drum.
- Stop parboiling when rice husks start to split open.
- Chalky grains or white centers indicate incomplete parboiling, which may cause breakage of grains during milling.
- Parboiling can be done in earthen pots or empty petrol drums depending on the quantity of rice.

- Note: steaming dexterises the kernels and drives the vitamin thiamine and other water-soluble nutrients from the testa or seed coat into the starchy core.

Transportation

In most developing and least developed countries like Nigeria that produce rice, conveying harvested paddy from the field to processing, marketing or storage areas are mainly by means of human and animal power and sometimes mechanical power with the corresponding devices, tools and equipment. Small and family-sized volume of paddy is transported in bags from the house storage to the small rice mill by foot, bullock carts, bicycles, motorcycles, small-sized vehicles, or public transport vehicles whichever mode is available and affordable. Where paddy is field threshed, men usually carry the 40-75kg bag of paddy on their shoulders or back from the field to the nearest road.

Paddy Storage

1. Clean grains by winnowing before storage
2. Store paddy at about 13% or less moisture content
3. Use bags or other small storage facilities.
4. Avoid moisture fluctuation, wetting and drying for good grain quality.
5. Store in dry sheltered condition.

What to note

1. The warehouse should be properly treated and disinfected before introducing the paddy.
2. Storage room should be well ventilated with windows facing north to south as this will avoid direct penetration of sunlight at dawn and early evening.
3. Always place paddy bags on a wooden pallets and place pallet at a 40-50 cm from the wall.
4. Warehouse window should be left open, and therefore it is advised that the windows be located slightly higher above the normal window position.
5. When several batches of paddy are stored, they should be removed on the principle of first in first out.
6. During long storage, storeroom should be disinfected with storage chemicals or other appropriate physical means.

Summary

1. Rice is mature when 80% of grains turned straw colour.
2. Delay in harvesting will cause the grains at the tip of shatter.
3. Threshing is done using legs to march on the straw
4. The major objective of rice drying is to reduce grain moisture content to prevent spoilage

Question and Answer Session

1. When is the best time to harvest rice?
2. How do you know when to harvest?
3. What is threshing?
4. Why do you dry rice?
5. What is parboiling?

Module 10
Record Keeping, Budget Planning and Decision making

Record keeping

Record keeping is the art of collecting useful pieces of data or information on the happenings of a particular undertaking, with the view of processing it in the future (for example, analysing sales and costs and calculating profits). Although record keeping is an important aspect of the family farm business, many farmers do not keep records. This makes it difficult for them to know how much they spent and how much income was earned from their farm business enterprises. Without good farm records, it becomes difficult for farmers to identify problem areas and know whether their businesses are generating profit or not. Therefore, record keeping is an important activity that is necessary for operating farm businesses
successfully. Family farm businesses should keep records, and this responsibility should be shared by all family members. Responsibility for any type of record should be allocated to the family member who is most capable, regardless of their sex.

Why should records be kept in a farm business?

- To track individual enterprise performance.
- To evaluate farm/enterprise profitability.
- To establish a basis for pricing.
- To facilitate access to loans.
- Records are helpful when farmers seek advice from extension agents.
- Records are a good management tool because they provide figures for planning and budgeting.

What is farm budgeting?

A budget is a formal plan for carrying out business activities in the future. It shows the process of carrying out an activity and the end result. Budgeting is the planning process or the development of a plan of action (budget). Families involved with farm businesses should budget together and share responsibilities in the budgeting process.

Uses of farm budgets

Budgets are normally used to:

- Decide what to produce, how much to produce, and the resources needed.
- Itemise the financial aspects of the farming business.
- List the inputs and production practices required by an enterprise.
- Evaluate the performance of different farm enterprises.
- Estimate benefits and costs of changes in production practices.
- Provide the basis for a total farm plan.
- Show the capacity of the farming business to carry risk.
- Support applications for credit (show the ability of the business to pay debt).
- Inform all interested parties of the costs incurred in producing an agricultural product\
-

Decision making

Decision-making is central to farm management. Each decision has an impact on the farm and on the farm household. Even deciding to do nothing is a decision and has an impact. The more a farmer is aware of the decision-making processes that affect farm and household, the more sustainable the enterprise will be and the more likely it will be profitable and sustainable.

Two of the main features for understanding economic decision-making are:

- The way farm boundaries (decision-making boundaries) are determined.
- The ultimate social objectives for which farm goods are produced.

Production decisions depend on a number of factors that fall outside the physical/ production area of the farm. One of these factors is the farm family/household. In many parts of Nigeria, the farm and the household are virtually one entity. Decisions about the farm directly impact on the household and decisions about the family directly impact on the farm.

Farm boundaries are determined by family structure.

There may be a family farm under the head of household, but the farm may have various sub-units over which a family member will have some level of control. This will have implications for decision-making, particularly with reference to shared labour and equipment. Traditionally men are heads of households, but there are now many variations. Even within a clearly established arrangement, different members of the family, especially women make different decisions for the family at different times of the year.

Key aspects of decision-making

In order to make fundamental decisions farmers need to develop understanding and skills in four broad decision areas:

- **Diagnosis:** Diagnosis means looking at the farm and household as it functioned over some period of time. For a first diagnosis, the farmer may want to understand how the farm has produced over several seasons.
- **Planning:** Once diagnosis is complete, planning can begin; decisions about what, how and how much to produce. Some planning decisions will be based on knowledge, such as how much land and labour are available. Other planning decisions will be based on less certain things, such as rainfall and product prices. The plan will also include an indication of the expected results in terms of yields and income.
- **Implementing**: Implementing means putting the plan into action. In general, implementing does not require major decisions. Things may not always work according to plan; less rainfall than expected, the price of a crop changes etc. During implementation, plans may need to be adjusted to accommodate such changes.
- **Monitoring & Evaluation:** Monitoring means keeping track of what is happening on the farm and the plan is a guide for monitoring. Monitoring provides the farmer with the information needed to evaluate the success of the plan.

Learning how to make decisions in these four areas will put farmers in greater command of the resources and processes that influence their food security and their income generation.

The four areas flow in a pattern which supports continuous learning processes about what works best for the farmer and the farm family.

Planting record sheet

Plant Date	Crop	Variety And the source	Total square meter planted
1/30/12	Rice		10000

Monthly Labour/activities records

Date:	Crop:	Bed Number(s):	Activity:	Time (to the15 minute):	Paid labor/unpaid (e.g. Yourself)
EXAMPLE	Rice	12	weeding	15 minutes	

	Date	Crop	Harvest amount (ton)	Quantity/ hectare	Quantity sold	Price	Total revenue	NOTES:
Exp:	5/1	Rice	24	12ton/ha	6 ton			
1)								
2)								
3)								
4)								
5)								
6)								
7)								
8)								
9)								
10)								
11)								
12)								
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Monthly/Revenue record sheet

Keep careful track each month of amounts harvested & how much revenue your selected crops bring you. Use a separate line for each crop and for each harvest date. Use additional sheets if necessary.